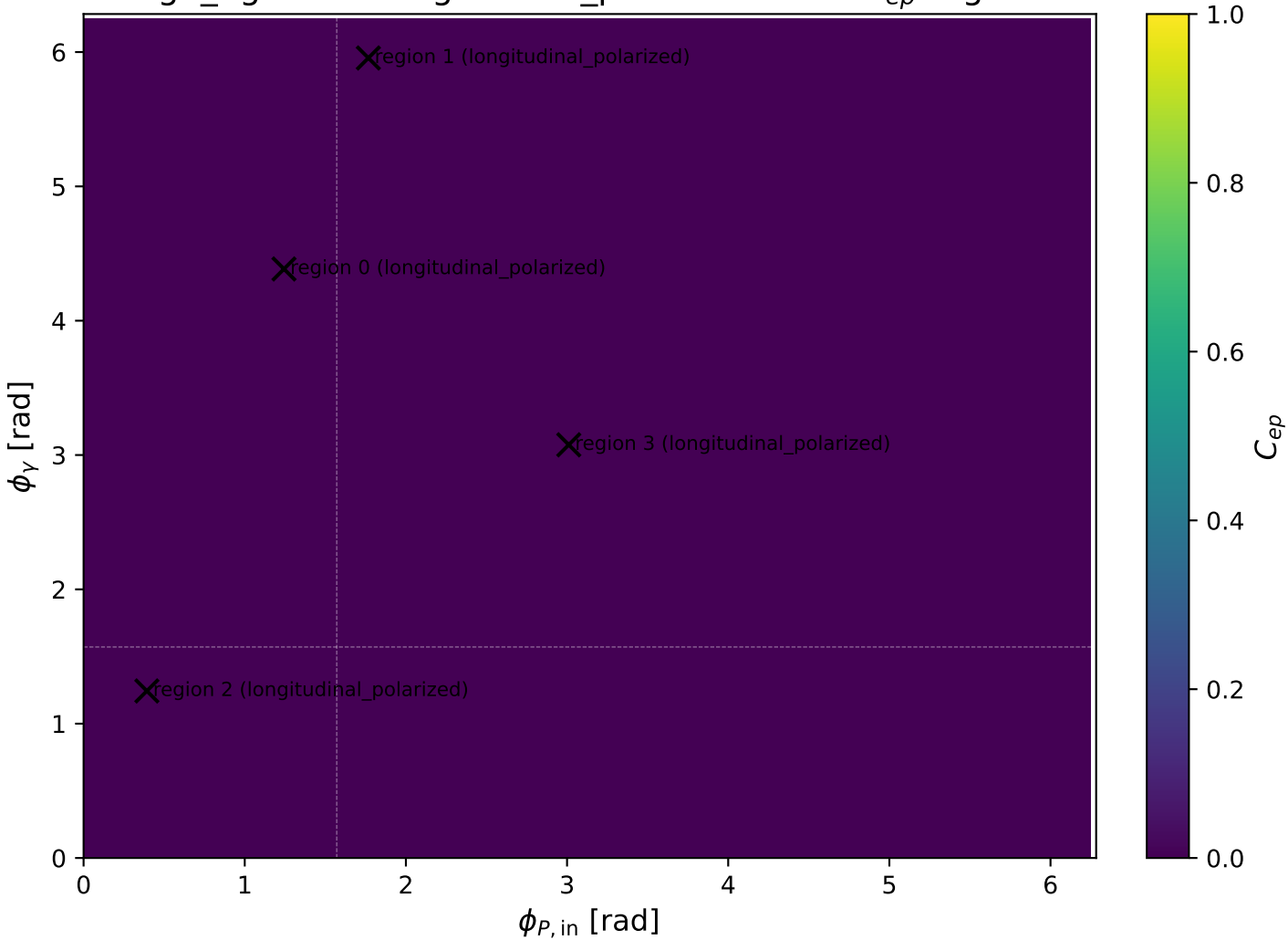
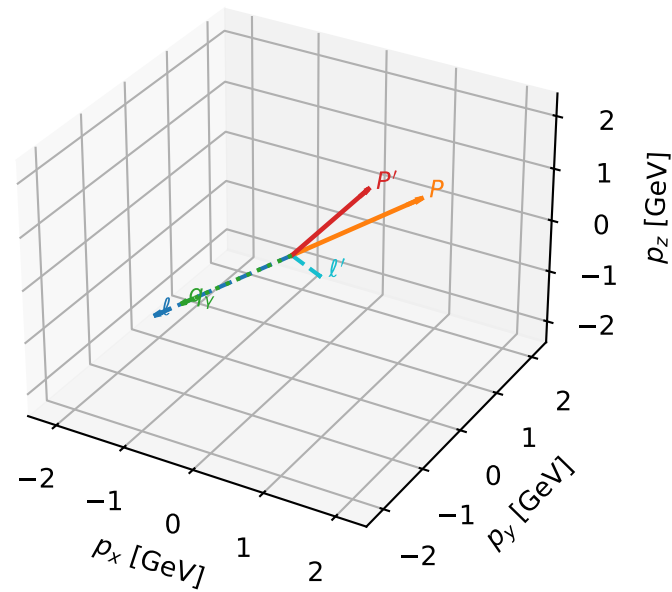


high\_Egamma longitudinal\_polarized: max  $C_{ep}$  regions



longitudinal\_polarized characteristic max C<sub>ep</sub> configuration

3D momenta

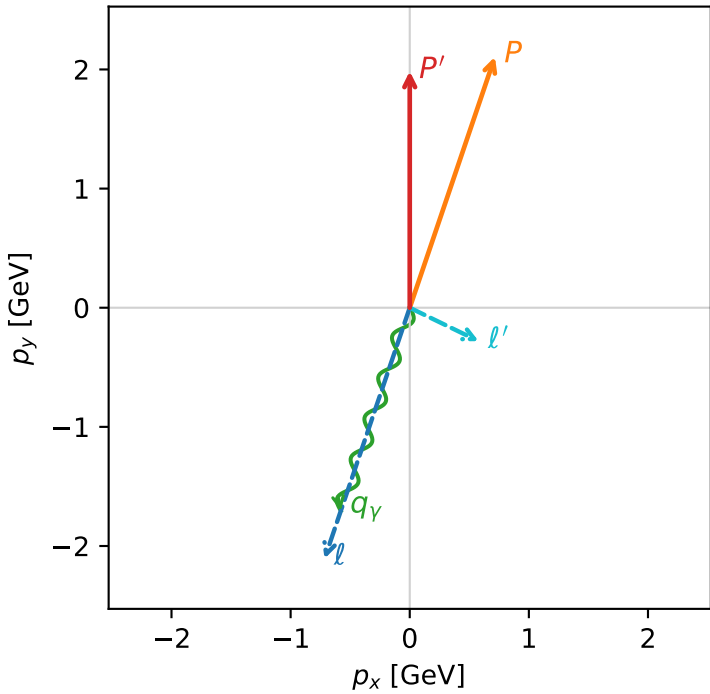


c\_ep\_high\_egamma\_longitudinal\_polarized\_region\_0 C\_{ep}=2.72251e-12  
region: high\_Egamma\_longitudinal\_polarized\_region\_0 final-pair delta\_xy=2.02229 rad  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $[A(h_e = +1, h_p = +1) - A(h_e = -1, h_p = +1)]/\sqrt{2}$

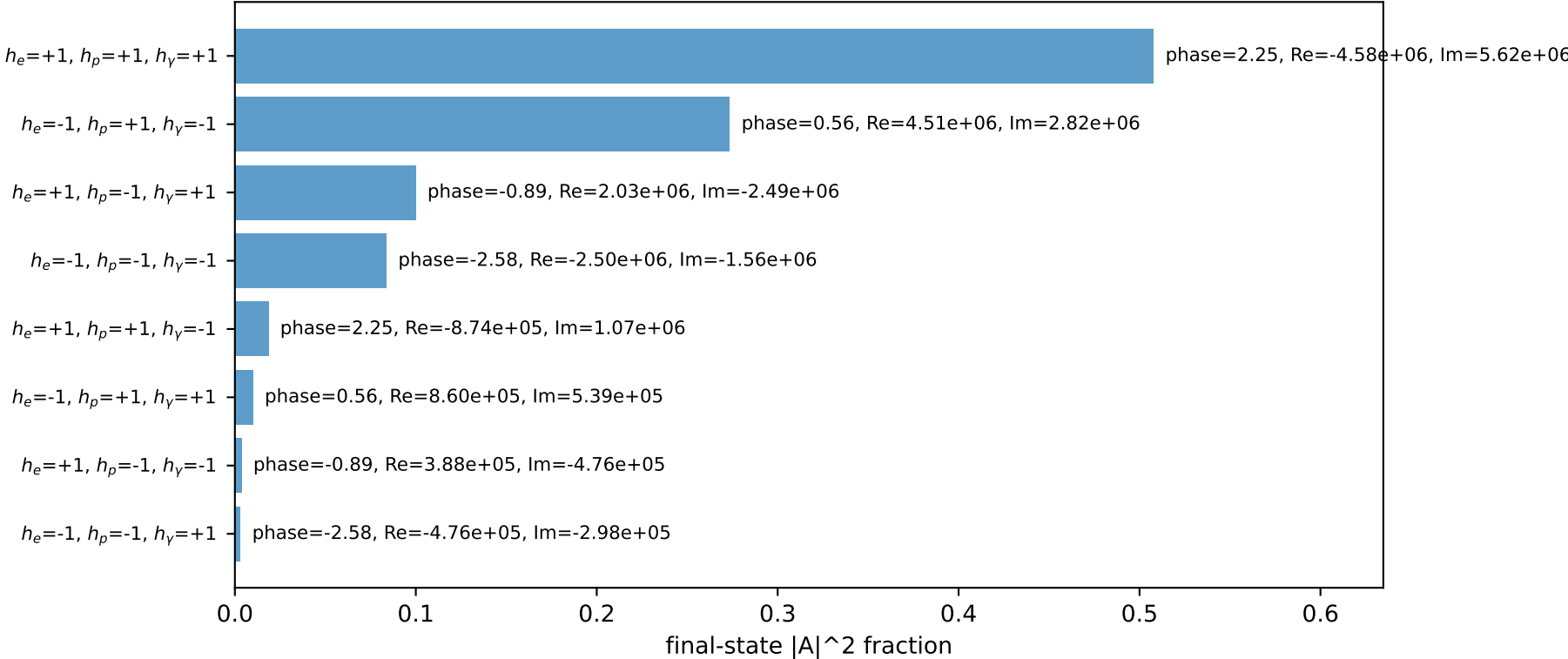
s=21.5158,  $\sqrt{s}$ =4.63852,  $|\vec{P}|$ =2.22442,  $|\vec{P}'|$ =1.98503  
 $\theta_{in}$ =1.57079,  $\phi_l$ =4.38514,  $\phi_p$ =1.24355  
 $E_\gamma$ =1.8,  $\phi_\gamma$ =4.38514  
 $Q^2$ =2.5062,  $x_B$ =0.145906,  $t$ =-0.478181,  $W^2$ =15.5505,  $y$ =0.832373  
 $\theta(l', \gamma)$ =82.8814 deg

four-momenta [E, p<sub>x</sub>, p<sub>y</sub>, p<sub>z</sub>] GeV:  
l E= 2.2244 p<sub>x</sub>= -0.7150 p<sub>y</sub>= -2.1064 p<sub>z</sub>= -0.0000  
P E= 2.4141 p<sub>x</sub>= 0.7150 p<sub>y</sub>= 2.1064 p<sub>z</sub>= 0.0000  
l' E= 0.6430 p<sub>x</sub>= 0.5786 p<sub>y</sub>= -0.2806 p<sub>z</sub>= 0.0000  
P' E= 2.1955 p<sub>x</sub>= 0.0000 p<sub>y</sub>= 1.9850 p<sub>z</sub>= 0.0000  
q<sub>γ</sub> E= 1.8000 p<sub>x</sub>= -0.5786 p<sub>y</sub>= -1.7045 p<sub>z</sub>= 0.0000

Transverse plane

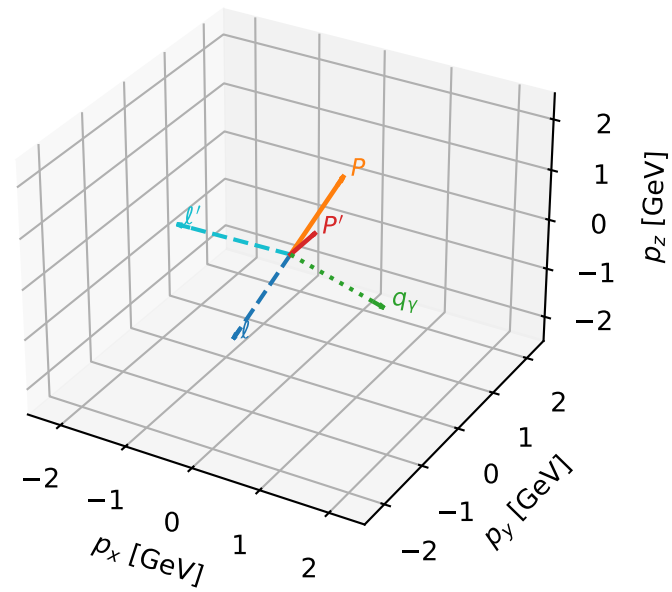


Final-state amplitude decomposition



longitudinal\_polarized characteristic max C<sub>ep</sub> configuration

3D momenta

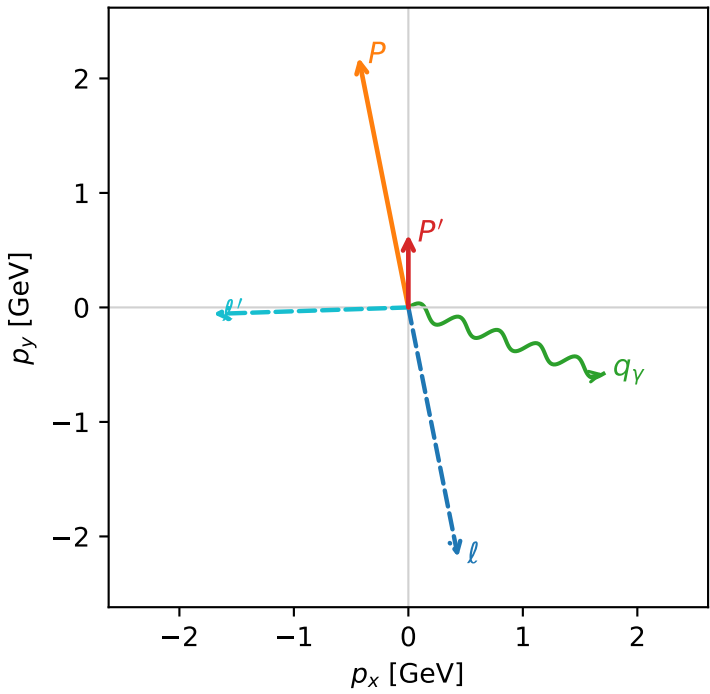


c\_ep\_high\_egamma\_longitudinal\_polarized\_region\_1 C<sub>ep</sub>=4.92611e-17  
region: high\_Egamma\_longitudinal\_polarized\_region\_1 final-pair delta\_xy=1.60426 rad  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $[A(h_e = +1, h_p = +1) - A(h_e = -1, h_p = +1)]/\sqrt{2}$

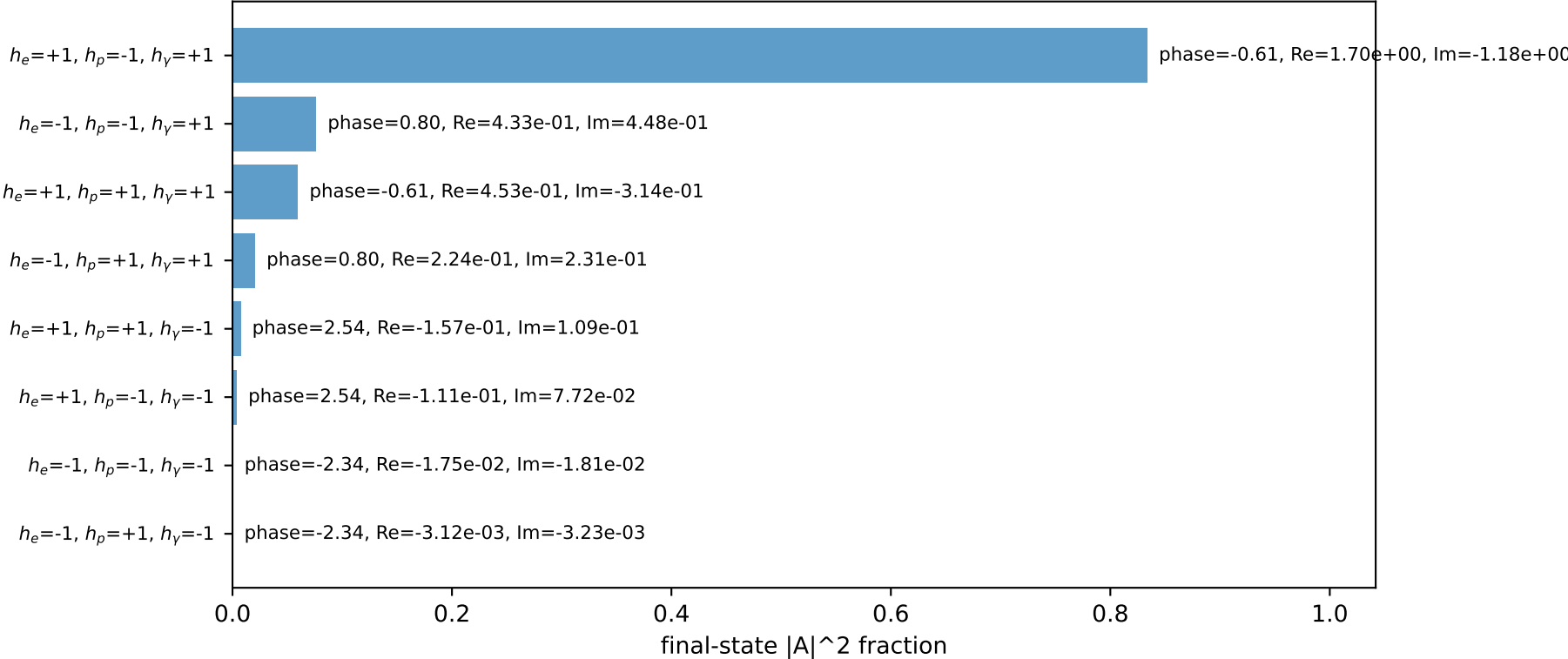
s=21.5158,  $\sqrt{s}$ =4.63852,  $|\vec{P}|$ =2.22442,  $|\vec{P}'|$ =0.635646  
 $\theta_{in}$ =1.57079,  $\phi_l$ =4.90874,  $\phi_p$ =1.76715  
 $E_\gamma$ =1.8,  $\phi_\gamma$ =5.95594  
 $Q^2$ =8.81758,  $x_B$ =0.646817,  $t$ =-0.937543,  $W^2$ =5.69452,  $y$ =0.660605  
 $\theta(l', \gamma)$ =159.333 deg

four-momenta [E, p<sub>x</sub>, p<sub>y</sub>, p<sub>z</sub>] GeV:  
l E= 2.2244 p<sub>x</sub>= 0.4340 p<sub>y</sub>= -2.1817 p<sub>z</sub>= -0.0000  
P E= 2.4141 p<sub>x</sub>= -0.4340 p<sub>y</sub>= 2.1817 p<sub>z</sub>= 0.0000  
l' E= 1.7054 p<sub>x</sub>= -1.7045 p<sub>y</sub>= -0.0571 p<sub>z</sub>= 0.0000  
P' E= 1.1331 p<sub>x</sub>= 0.0000 p<sub>y</sub>= 0.6356 p<sub>z</sub>= 0.0000  
q<sub>γ</sub> E= 1.8000 p<sub>x</sub>= 1.7045 p<sub>y</sub>= -0.5786 p<sub>z</sub>= 0.0000

Transverse plane

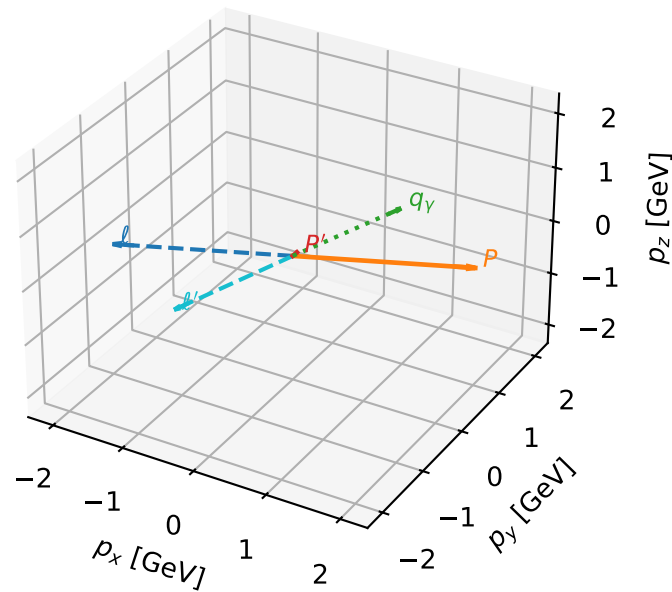


Final-state amplitude decomposition



longitudinal\_polarized characteristic max C<sub>ep</sub> configuration

3D momenta

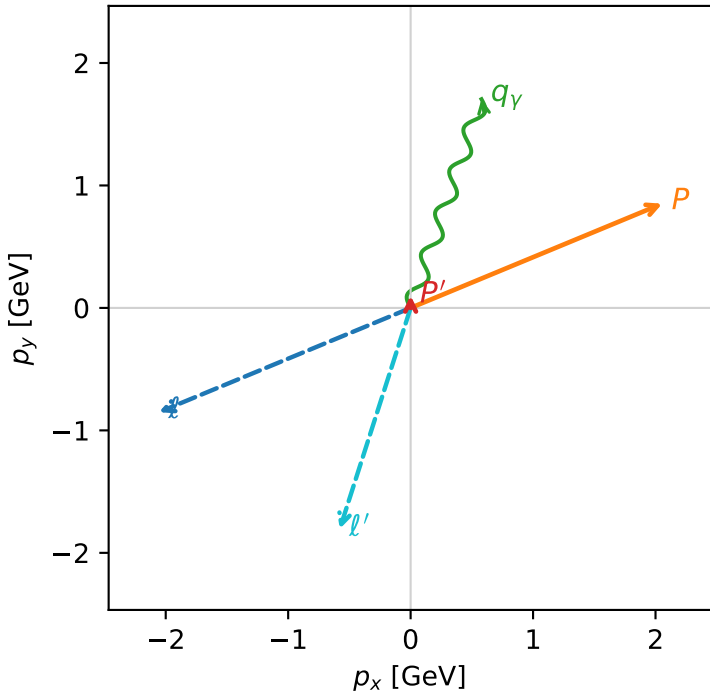


c\_ep\_high\_egamma\_longitudinal\_polarized\_region\_2 C<sub>ep</sub>=2.83708e-17  
region: high\_Egamma\_longitudinal\_polarized\_region\_2 final-pair delta\_xy=2.83134 rad  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $[A(h_e = +1, h_p = +1) - A(h_e = -1, h_p = +1)]/\sqrt{2}$

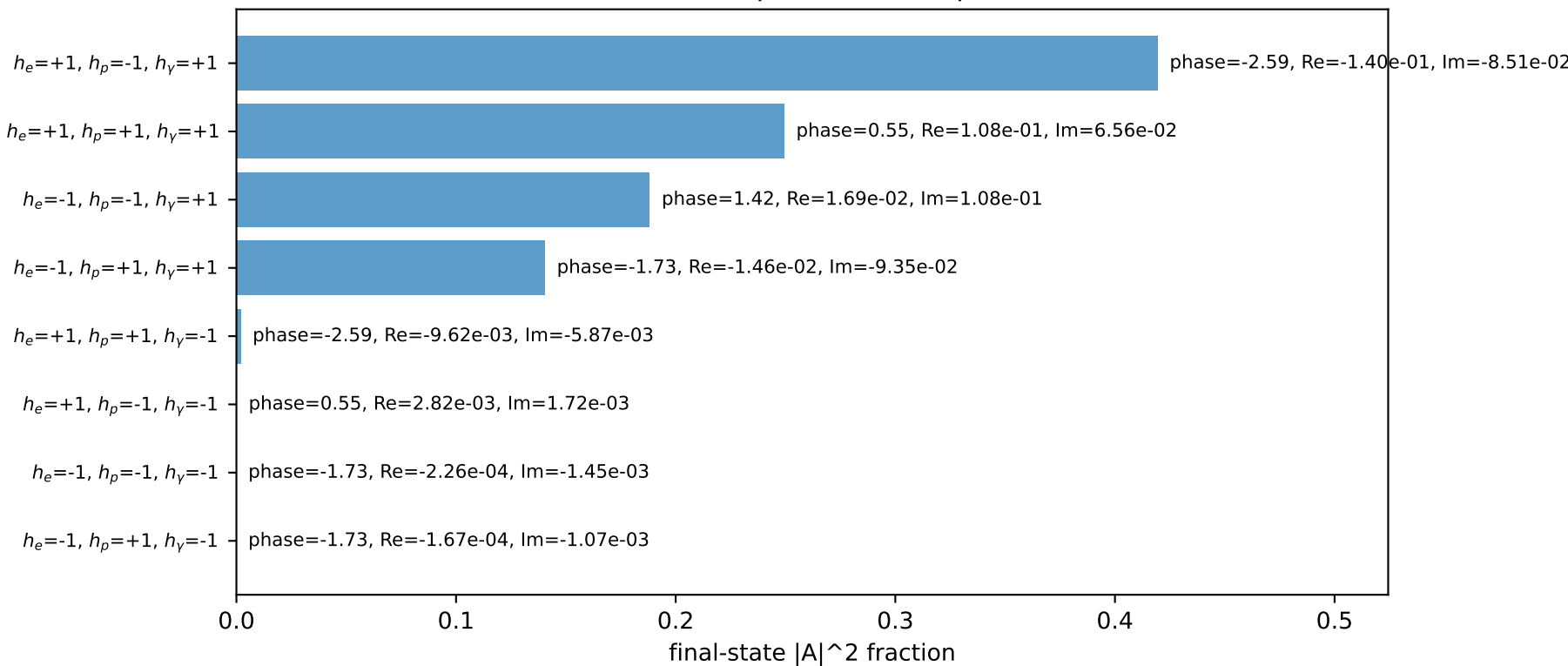
s=21.5158,  $\sqrt{s}$ =4.63852,  $|\vec{P}|$ =2.22442,  $|\vec{P}'|$ =0.100223  
 $\theta_{in}$ =1.57079,  $\phi_t$ =3.53429,  $\phi_P$ =0.392699  
 $E_\gamma$ =1.8,  $\phi_\gamma$ =1.24355  
 $Q^2$ =2.98073,  $x_B$ =0.493899,  $t$ =-2.62431,  $W^2$ =3.93421,  $y$ =0.292455  
 $\theta(\ell', \gamma)$ =179.026 deg

four-momenta [E, p<sub>x</sub>, p<sub>y</sub>, p<sub>z</sub>] GeV:  
 $\ell$  E= 2.2244 p<sub>x</sub>= -2.0551 p<sub>y</sub>= -0.8512 p<sub>z</sub>= -0.0000  
 $P$  E= 2.4141 p<sub>x</sub>= 2.0551 p<sub>y</sub>= 0.8512 p<sub>z</sub>= 0.0000  
 $\ell'$  E= 1.8952 p<sub>x</sub>= -0.5786 p<sub>y</sub>= -1.8047 p<sub>z</sub>= 0.0000  
 $P'$  E= 0.9433 p<sub>x</sub>= 0.0000 p<sub>y</sub>= 0.1002 p<sub>z</sub>= 0.0000  
 $q_\gamma$  E= 1.8000 p<sub>x</sub>= 0.5786 p<sub>y</sub>= 1.7045 p<sub>z</sub>= 0.0000

Transverse plane

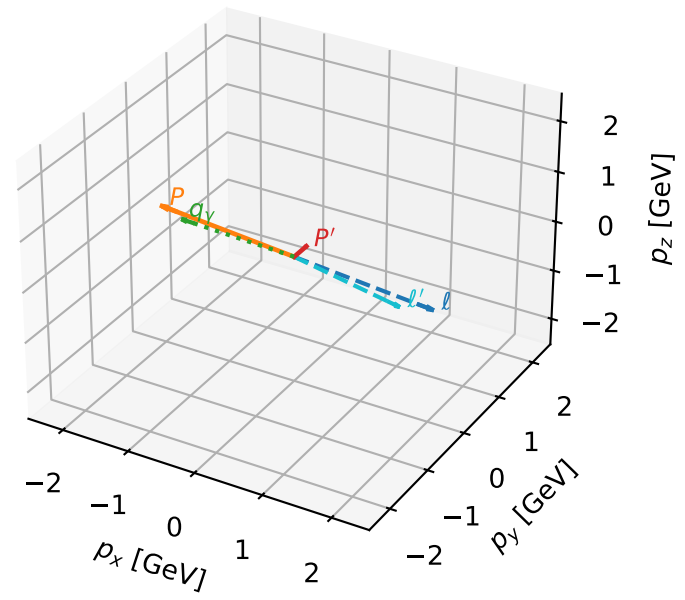


Final-state amplitude decomposition



# longitudinal\_polarized characteristic max $C_{ep}$ configuration

## 3D momenta

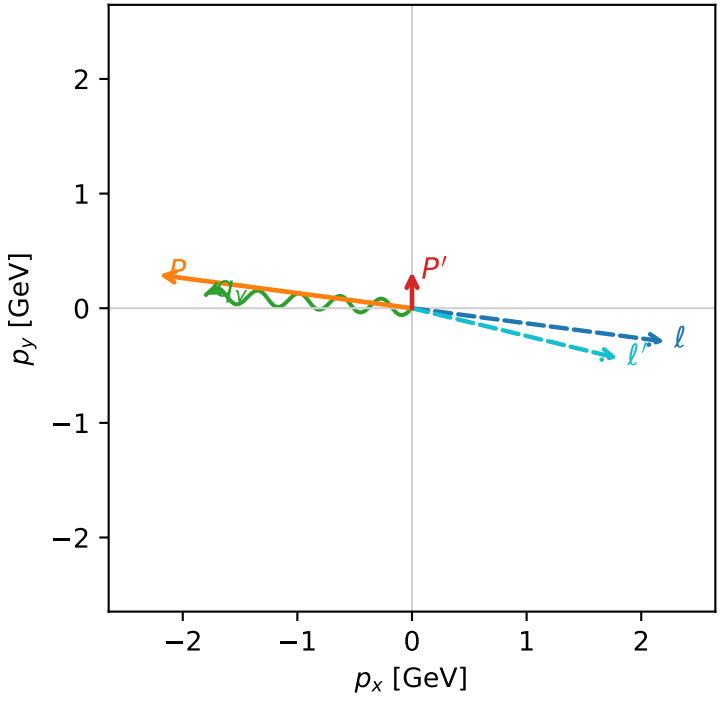


c\_ep\_high\_egamma\_longitudinal\_polarized\_region\_3  $C_{ep}=2.19187\text{e-}17$   
region: high\_Egamma\_longitudinal\_polarized\_region\_3 final-pair delta\_xy=1.80863 rad  
kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
incoming state:  $[A(h_e = +1, h_p = +1) - A(h_e = -1, h_p = +1)]/\sqrt{2}$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.317708$   
 $\theta_{in}=1.57079$ ,  $\phi_l=6.15229$ ,  $\phi_p=3.01069$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=3.07614$   
 $Q^2=0.0469691$ ,  $x_B=0.0132779$ ,  $t=-2.8374$ ,  $W^2=4.37028$ ,  $y=0.171419$   
 $\theta(l', \gamma)=170.123$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $l$   $E= 2.2244$   $p_x= 2.2054$   $p_y= -0.2903$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.2054$   $p_y= 0.2903$   $p_z= 0.0000$   
 $l'$   $E= 1.8482$   $p_x= 1.7961$   $p_y= -0.4354$   $p_z= 0.0000$   
 $P'$   $E= 0.9903$   $p_x= 0.0000$   $p_y= 0.3177$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -1.7961$   $p_y= 0.1177$   $p_z= 0.0000$

## Transverse plane



## Final-state amplitude decomposition

